Susceptibility and Spatial Signature of Abrupt Thaw Across Alaska's Permafrost Landscapes

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Key points

- Key point 1
- Key point 2
- Key point 3

Abstract

Abstract text body.

Plain Language Summary

Plain-language summary body.

1 Introduction

The northern permafrost region contains more organic carbon in its upper three meters of soil than currently resides in Earth's atmosphere (Hugelius et al., 2014). Most permafrost thaw proceeds by gradual, top-down deepening of the active layer, but abrupt thaw destabilizes and collapses the ground surface over days to years, mobilizing carbon stocks that gradual thaw would take decades to reach (Turetsky et al., 2020; Nitzbon et al., 2020). Under high emissions scenarios, abrupt thaw is projected to impact only 2.5 million km^2 out of the 18 million km^2 permafrost region, but threatens to release nearly as much carbon as gradual thaw processes operating over a much larger area (Turetsky et al., 2020). Abrupt thaw leads to a variety of landforms, including thermokarst lakes, collapsing peatlands, and hillslope failures, all of which differ in the quantity and form of carbon released (Olefeldt et al., 2016; Schädel et al., 2016). The scale and timing of carbon release from permafrost soils therefore depend on how and where the landscape undergoes abrupt thaw, and this spatial pattern remains one of the largest sources of uncertainty in the permafrost-carbon feedback (Turetsky et al., 2020; Walter Anthony et al., 2018).

Beyond climate feedbacks, abrupt thaw represents an immediate hazard to the roughly 5 million people who live in the Arctic circumpolar permafrost region (Ramage et al., 2021). Projections suggest that 30-50% of Arctic infrastructure may be at risk of damage by permafrost thaw by 2050, threatening tens of billions of U.S. dollars in damage by the end of the century (Hjort et al., 2022). In Alaska alone, permafrost thaw may cause \$37 to \$51 billion in damage to buildings and roads (Manos et al., 2025). This burden will fall disproportionately on remote and Indigenous communities, whose unofficial, "vernacular" infrastructure and assets often lie outside conventional damage inventories (Verstraete & Zachwieja, 2025). Between 1974 and 2019, thermokarst at Point Lay, Alaska, expanded nearly an order of magnitude faster beneath the developed village than in the surrounding tundra, indicating a potential positive feedback between the built environment and abrupt thaw (Jones et al., 2025). Predicting how and where abrupt thaw will develop on the landscape is therefore critical for helping Arctic communities adapt to or mitigate the effects of future climate change.

Abrupt permafrost thaw proceeds faster than gradual thaw, by definition, but also results from different processes, leaves distinct surface expressions, and has specific consequences for ecosystems, hydrology, and carbon release (Webb et al., 2025). Gradual thaw typically proceeds from the top down as a progressive deepening of the seasonally thawed active layer into near-surface permafrost (Walter Anthony et al., 2018; Webb et al., 2025). Abrupt thaw, by contrast, is not confined to this vertically ordered progression: subsidence, erosion, mass wasting, and the flow of water and heat can rapidly propagate thaw downward and laterally, affecting deeper or adjacent permafrost (Fig. 1) (Turetsky et al., 2020; Nitzbon et al., 2020; Webb et al., 2025). The resulting changes to the landscape occur quickly relative to geomorphic time scales, from vertical subsidence of almost a meter over a decade (Farquharson et al., 2019) to thaw-slump headwall retreat of more than ten meters in a single year (S. Kokelj et al., 2015). A site's thaw mode, abrupt or gradual, is not dictated by the presence of permafrost alone, but emerges from an interplay of ground ice, geomorphology, hydrology, and disturbance history (Jorgenson & Osterkamp, 2005; S. V. Kokelj & Jorgenson, 2013).

Permafrost has been mapped extensively across the Arctic, but existing products largely resolve conditions other than the mode of thaw. Many characterize the substrate, outlining permafrost extent and zonation (Brown et al., 1997), ground ice content (Jorgenson et al., 2008), and probability of occurrence (Obu et al., 2019). Others document where abrupt thaw features have formed, such as ponds and lakes (Muster et al., 2017) or retrogressive thaw slumps (Nitze et al., 2025), or quantify how far degradation of specific features has advanced (Zhang et al., 2025). A growing set of susceptibility maps predicts where specific abrupt thaw processes may develop, including thermokarst lakes (Wang et al., 2023), retrogressive thaw slumps (Makopoulou et al., 2024), and active layer detachment (Makopoulou et al., 2025), and a recent hemispheric model combines lake and slump susceptibility into a single product (Yang et al., 2025). Each of these efforts estimates the likelihood that a named landform will occur at any given point, and the combined model remains a union of predictions anchored to specific processes. None of these previous efforts addresses the question of thaw mode: whether the conditions at a site resemble those under which abrupt or gradual thaw has been observed, independent of any single landform (Fig. 1). Across Alaska, that contrast is carried only by point-wise samples from expert analysis of field and remote sensing observations (Webb et al., 2026), and not yet mapped as a continuous, predictive surface.

Here, we map thaw mode as a continuous, predictive surface across Alaska. We train a gradient-boosted classifier on 19,288 expert-labeled thaw points from the Alaska Permafrost Thaw Database (v2.0) (Webb et al., 2026). The classifier evaluates sites along a gradient of how strongly key environmental variables resemble either abrupt or non-abrupt thaw, using 70 geospatial indicators sourced from remote sensing, community data products, and model reanalysis. To mitigate sampling bias inherent to the Thaw Database, we evaluate the model under spatial cross-validation, bounded by an applicability mask that indicates where points fall within the envelope of the training data, and then apply it to a statewide feature grid populated by the same data sources. The resulting product is a prior-free, log-evidence index of thaw mode. We then use grouped Shapley (SHAP) values (Lundberg & Lee, 2017; Lundberg et al., 2020) to identify which geospatial indicators best distinguish between thaw modes and how they shift the classifier's prediction. This paper describes the expert-labeled points and feature stack in Section 2, model construction and evaluation in Section 3, the susceptibility map and its drivers in Section 4, and limitations, interpretations, and comparisons with the broader permafrost mapping literature in Section 5.

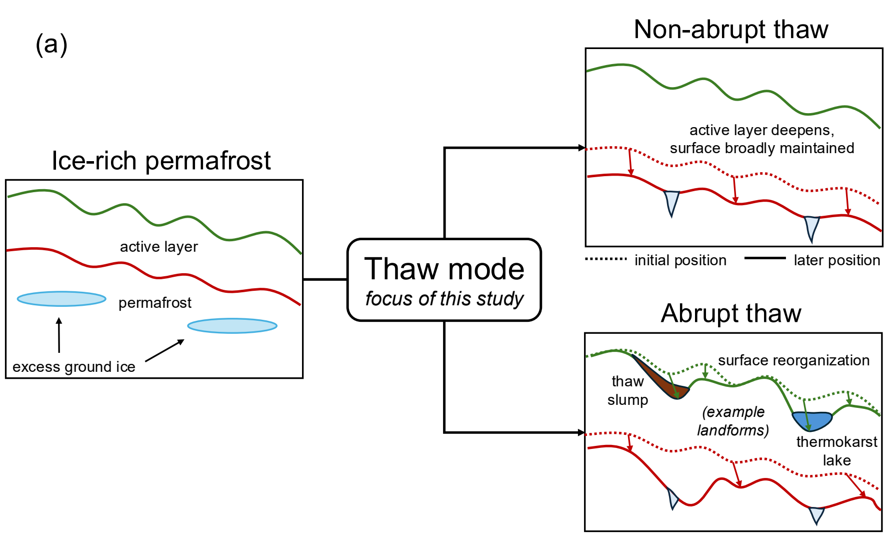


Figure 1.

2 Data

2.1 Labeled thaw points

We train and evaluate the classifier on expert-labeled observations of thaw mode from the Alaska Permafrost Thaw Database (v2.0) (Webb et al., 2026). The Thaw Database compiles observations of permafrost across Alaska from a wide variety of sources, including both field and remote-sensing studies. After removing exact duplicates in feature space, the training set comprises 19,288 points, each assigned a binary thaw mode label. We encode abrupt thaw as `class 0`, and then all non-abrupt thaw labels as `class 1`. Abrupt thaw dominates the dataset with 18,181 points (94.26%), alongside only 1,107 (5.74%) non-abrupt thaw points. Notably, this composition is the inverse of the landscape it samples. The majority of observations in the Thaw Database are `class 1`, yet abrupt thaw is projected to impact only a fraction of the permafrost region overall (Turetsky et al., 2020).

The non-abrupt thaw label denotes sites undergoing gradual rather than abrupt thaw. Roughly 18% of these points are drawn from the long-term permafrost monitoring networks (the Global Terrestrial Network for Permafrost boreholes and Circumpolar Active Layer Monitoring program). We label these points as “non-abrupt” rather than “gradual” thaw deliberately, following the convention in the Thaw Database, to include sites where thaw is not definitively abrupt. The abrupt thaw class contains a variety of landforms, including thermokarst lakes, collapsing peatlands, retrogressive thaw slumps, and active-layer detachments, among others. We refer the reader to Webb et al. (2026) for the full decision matrix used to label points.

Because the observations were largely assembled from field campaigns and targeted remote-sensing surveys, the spatial distribution of labeled points reflects the research focus of the contributing studies and physical accessibility of sites. Figure 2 shows the distribution of abrupt (a) and non-abrupt (b) thaw observations. The two classes are not evenly balanced across the landscape, with many more abrupt thaw studies focused in the north, and non-abrupt sites spread more evenly across the state. This uneven coverage motivates the spatial cross-validation and area-of-applicability analyses described in Section 3. We carry site coordinates through the pipeline as metadata and use them to build cross-validation blocks, but exclude them from the feature table so that the model cannot use location itself as a predictor.

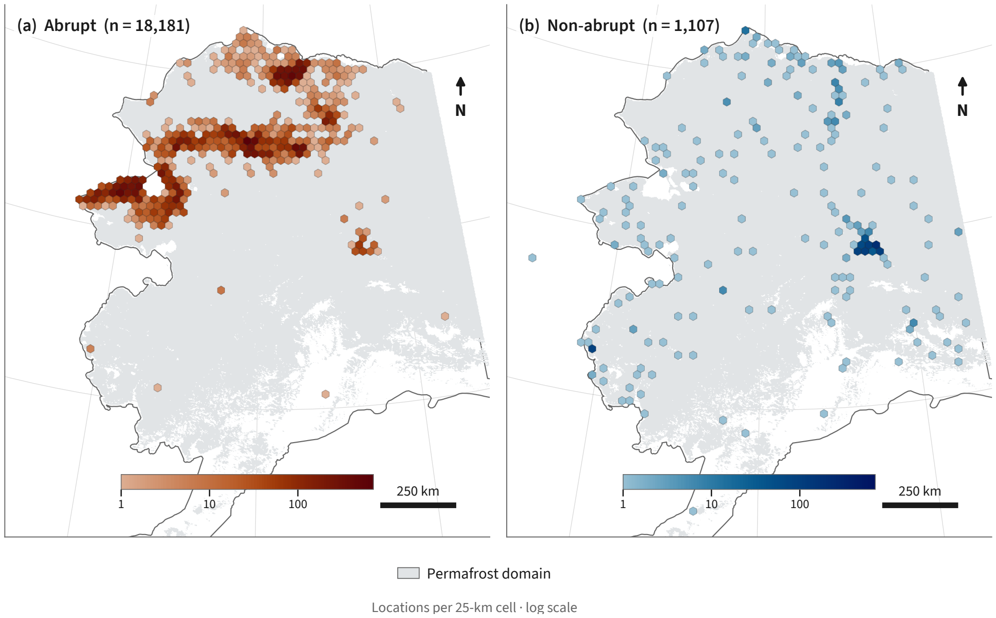


Figure 2.

2.2 Geospatial features

Each point is characterized by a common set of geospatial features extracted at its location, which together form a feature table of 19,288 rows and 70 columns. Continuous variables enter the matrix directly, while categorical variables are one-hot encoded into individual columns. Data sources include remote sensing products, modeled or interpolated data products, and gridded climate products (Table A1).

We obtain surface topography from the USGS 3D Elevation Program 10 m digital elevation model (U.S. Geological Survey, 2019). We use elevation, slope, aspect (decomposed into northness and eastness), and mean curvature computed at 500 m and 2 km scales. We account for surface hydrology using MERIT Hydro (Yamazaki et al., 2019), pulling height above nearest drainage and upstream contributing area. Land cover classifications come from the National Land Cover Database for Alaska (Dewitz, 2019), and are one-hot encoded into 18 indicator columns. Fire history is derived from the MODIS MCD64A1 burned-area product (Giglio et al., 2018), expressed as time since last fire and cumulative burn count.

We obtain soil properties from SoilGrids (Poggio et al., 2021), including organic carbon, total nitrogen, bulk density, and soil texture in sand, silt, and clay fractions. To avoid collinear columns (where sand, silt, and clay fraction sum to a fixed total), we encode soil texture as sand fraction and clay fraction, dropping silt from the feature table. We then aggregate the soil data into two depths, "surface" from 0-30 cm and "depth" from 30-200 cm, to aid in later interpretability. We include relative flammability as a continuous column and dominant vegetation mode encoded into 7 indicator columns, both sourced from the ALFRESCO landscape succession model (Bennett et al., 2017). We include a binary classification of yedoma from the Ice-Rich Yedoma Permafrost Inventory (Strauss et al., 2022), denoting points that are classified as "confirmed" yedoma in the IRYP inventory. Nineteen bioclimatic variables describe annual and seasonal temperature and precipitation, sourced from the WorldClim v1 climatology (Hijmans et al., 2005). Mean annual snow water equivalent (SWE) and multidecadal trends in SWE, precipitation, and temperature come from Daymet v4 (Thornton et al., 2022), computed from 1991-2020.

Across all data products, we do not impute missing values, choosing instead to select a classification algorithm that handles NaN values natively (Section 3). Two sources have structured gaps worth noting: SoilGrids is missing at approximately 16% of our training points, and the MODIS fire domain thins considerably above 70°N, leaving the northern Arctic coast without fire history coverage. Table A1 lists all data products with their native resolutions and references.

2.3 Prediction datacube

Once we have a trained classifier, producing predictions of thaw mode across Alaska requires mapping the same 70 geospatial indicators onto a statewide grid. We build the prediction datacube at a nominal resolution of 1 km on a fixed 0.00898° grid in geographic coordinates (EPSG:4326). True cell area varies with latitude, narrowing towards the north. We sample each data product onto the grid with the exact same extraction and post-processing used for the feature table, described above. Continuous rasters are resampled from their native resolution to the grid resolution, categorical layers are sampled by nearest neighbor, and yedoma is assigned by polygon membership. This parity ensures that the trained model transfers from the training points to the statewide grid without a train/serve mismatch. We restrict all predictions to the permafrost domain defined by the Obu et al. (2019) permafrost probability map, where the classification of thaw mode is physically meaningful.

3 Methods

To map thaw mode across Alaska, we train a gradient-boosted classifier on the labeled thaw points, evaluate its skill with spatial cross-validation, and bound it with an area-of-applicability analysis. Our result is a log-evidence index of thaw mode, which we then map statewide by applying the trained model. Our methods account for the spatial clustering of abrupt thaw points, the relative sparsity of non-abrupt thaw points in the training dataset, and the lack of expert-verified thaw observations from vast regions within the state. After training and deploying the model, we then apply post-hoc interpretability tools to understand the features that drive the model and produce the strongest indications of abrupt thaw.

3.1 Gradient-boosted classification

We use gradient-boosted decision trees to classify thaw mode, following the XGBoost algorithm (Chen & Guestrin, 2016). XGBoost has two key advantages for our problem. First, several of our features have missing values that are structured, rather than simply denoting a random point of missing data. For example, SoilGrids does not carry soil data over open water features, and similarly, the MODIS fire data is not available at very high latitudes. Gradient-boosted trees learn directly from incomplete rows, and do not require us to invent a structure for missing values or handle them ad hoc. Second, XGBoost allows for exact per-feature attribution through the TreeSHAP algorithm (Lundberg et al., 2020), which we use to build model interpretability (§3.4).

We measure model skill as the area under the precision-recall curve, as appropriate for a rare positive class (i.e., non-abrupt thaw here) (Saito & Rehmsmeier, 2015). Overall accuracy and the area under the receiver operating characteristic (ROC) curve are both dominated by abrupt thaw, and can appear high even when the model fails to identify non-abrupt sites. We report a single ROC value for comparison with other work, but otherwise evaluate the model through precision-recall. We select hyperparameters to maximize this score (Appendix B) using nested cross-validation and then refit the final model on all labeled points. Rather than reweight the classes (abrupt and non-abrupt) to compensate for their relative prevalence in the database, we keep the sample prior intact, allowing us to later produce a prior-free index of thaw mode (§3.3). We also fit a logistic regression model to the data to act as a baseline model, but we do not test our pipeline against any other machine learning algorithms.

3.2 Spatial evaluation

Figure 2 shows the spatial clustering of abrupt and non-abrupt thaw points from the training dataset. The underlying variables that we use to predict thaw mode vary smoothly in space, following Tobler's first law of geography that "near things are more related than distant things" (Tobler, 1970). Therefore, a random test-train split on these data would place near-duplicate neighbors on either side of the split, inflating the model's apparent skill.

To avoid this bias, we evaluate the model using spatial cross-validation. We divide the state into square blocks on an equal-area grid and hold out entire blocks together, so that clusters of nearby points fall on the same side of the test-train split, rather than across it. The appropriate block size is not obvious, so we repeat the evaluation on blocks ranging from 10 to 200 km on a side. Larger block sizes push the test points further away from the training data, so this systematic approach gives one measure of how well the model generalizes as we increase the distance away from its training data. We do not use an additional buffer zone around the blocks, as a sweep of buffer sizes indicated that this was irrelevant for leakage (Appendix C).

Block cross-validation tests the model's ability to generalize beyond immediate neighborhoods, but does not scale up to a full statewide analysis. To probe statewide generalization, we also withhold entire geographic regions in turn and measure how skill degrades as the model is forced to predict on progressively less sampled terrain. We repeat the block-to-fold assignment many times to confirm that measured skill does not depend on any single partition.

3.3 Mapping thaw mode

Applying the trained model to our statewide feature grid yields an estimated probability that any given cell exhibits abrupt thaw. However, we do not report this probability directly. The model is calibrated to the class balance in the training database, and that balance reflects how the observations were collected much more than the true proportion of abrupt thaw across Alaska. Prior work suggests abrupt thaw will affect a minority of the permafrost region (Turetsky et al., 2020), rather than the roughly 94% that random sampling of our database would imply. The true landscape proportion is not recoverable from a biased sample, so rather than use a calibrated probability, we instead build a prior-free index of evidence. In plain language, a high index value indicates high evidence that a grid cell matches the features associated with abrupt thaw.

For a cell with a feature vector $\vec{x}$, let $p(\vec{x})$ be the model's estimated probability of abrupt thaw, and then let Π be the prevalence of abrupt thaw in the training data. We then define the log-evidence index as:

$$s(\vec{x}) = \mathrm{logit}\big(p(\vec{x})\big) - \mathrm{logit}(\Pi), \tag{1}$$

where $\mathrm{logit}(z) = \log[z/(1-z)]$ maps a probability to a real number. This is the logarithm of the likelihood ratio between abrupt and non-abrupt thaw, as indicated by the local features at a given grid cell. A value of zero is neutral, while positive values favor abrupt thaw and negative values favor non-abrupt thaw. The index is neither a probability nor a discrete class, so we will call it a *log-evidence index* here.

Because we did not reweight the classes during training, the model holds the sample prevalence Π as its prior, and then subtracting $\mathrm{logit}(\Pi)$ removes it. This means that the resulting index does not depend on class balance at all. Given new observations with a significantly different ratio of abrupt to non-abrupt thaw points, $p(\vec{x})$ would change, but $s(\vec{x})$ would not, because the prior is removed from the index. This insulates the index from raw class balance concerns, but does not yet address sampling that is uneven across feature space.

The model will return a prediction at any grid cell, but the training dataset does not cover every combination of environmental variables found across the state of Alaska. Where a cell lies far away from the training data in feature space, the model extrapolates and its output should not be trusted. To delineate these regions, we adapt the area-of-applicability method from Meyer and Pebesma (2021). For each cell, we calculate a dissimilarity index, measuring the distance in feature space to the nearest training point, with features weighted by importance to the model and scaled by average dissimilarity among training points. We set the applicability threshold at the 99.9th percentile of this index measured across all of the training points, corresponding to a dissimilarity index of 0.27. Cells outside of that range are flagged as extrapolation. This threshold does not denote a limit of model skill, as we found that the model performance did not degrade as the dissimilarity index rose (Appendix D). Instead, the threshold is chosen as the reasonable extent of the feature space, beyond which skill cannot be measured regardless.

3.4 Interpretability

To interpret the model's classification decisions, we aim to attribute output values to input features. Many of the features are strongly correlated with one another, so a simple per-feature interpretation would risk diluting one physical signal across several environmental variables. As such, we group the features in the model first, before attributing importance, using the correlation structure of the features alone. To do so, we measure the correlation between every pair of features and join similar features into families by hierarchical clustering. Our categorical variables, which are encoded as one-hot columns, are collapsed into families by data source. Features that are strongly anti-correlated are treated as redundant as well, because a variable and its inverse carry the same information. The number of families is not fixed in advance, but emerges from the correlation structure (Appendix E).

With the feature families defined, we quantify each group's contribution to the model using SHAP values (Lundberg & Lee, 2017), computed for our tree model with the TreeSHAP algorithm (Lundberg et al., 2020). A SHAP value measures how far a feature moves the model's output for a single prediction, as compared to its average output. We sum the SHAP values within each family to obtain one contribution per family, and keep the same groupings throughout.

We compute two different sets of SHAP values, for two purposes. First, to rank the families by importance, we compute SHAP values on held-out points during spatial cross-validation, refitting the model within each fold so that every attribution is made on data the model did not see in training. This provides an unbiased ranking on held-out data. Second, to map how the drivers of abrupt thaw vary across the state, we compute SHAP values from the final, all-data model at every grid cell and record the family that dominates in each. This gives a spatially complete display, using the actual deployed model. Both computations share the same family definitions, so the ranking and the map remain consistent.

4 Results

4.1 Alaska's susceptibility to abrupt thaw

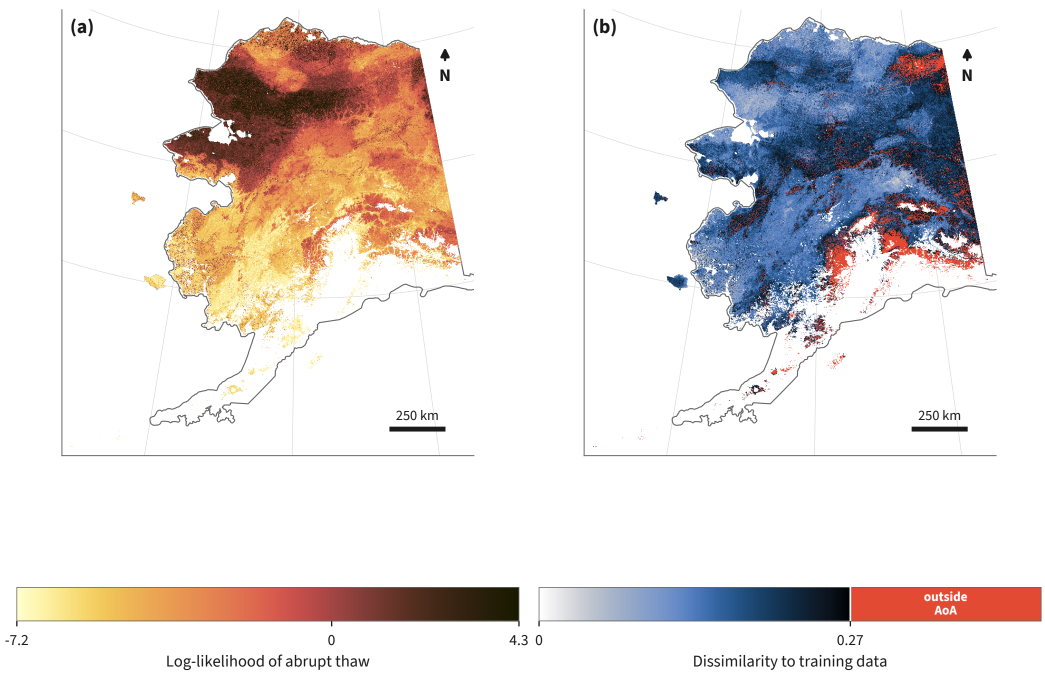


Figure 3.

Figure 3a maps the log-evidence index of thaw mode across Alaska's permafrost domain. Within the area of applicability (AoA), 24.7% of the landscape exhibits features that are more consistent with abrupt thaw than with non-abrupt thaw, corresponding to an area of roughly 264,000 km^2 out of the 1.07 million km^2 of the in-AoA permafrost domain. We denote a log-evidence of zero as the neutral boundary, where features favor neither abrupt nor non-abrupt thaw. The log-evidence map describes the current characteristics of the landscape, based on the input features and our model's interpretation, rather than a deterministic prediction of where abrupt thaw has occurred, is occurring, or will occur across the state.

Most of the domain is more consistent with non-abrupt thaw than abrupt thaw. The median in-AoA log-evidence is -2.46, so at the typical grid cell, the model favors the non-abrupt thaw mode by roughly a factor of 10. However, thaw mode is not evenly spread across the state. Abrupt-favored terrain concentrates in the tundra lowlands of western and northern Alaska, including the Seward Peninsula, the Brooks Range foothills, and the Arctic coastal plain. Conversely, the boreal interior and southern mountains primarily favor non-abrupt thaw. We investigate the spatial patterns of thaw mode in more detail in §4.3.

Figure 3b shows the dissimilarity index, which we use to determine the AoA for our model. Given the threshold set in §3.3, the mask flags 9.7% of the mapped domain as extrapolation beyond the envelope of the training features. The out-of-AoA cells are mostly due to features with long-tailed distributions, such as snow water equivalent (mean and trend), which accounts for nearly half of the dissimilarity score for flagged cells. Over the range we can measure, the model skill does not degrade with dissimilarity, so the bound of AoA shows where the input features are no longer represented by the training data, rather than where the model begins to fail.

4.2 Evaluating out-of-sample skill

Producing a trustworthy map of thaw mode depends entirely on the deployed model's ability to generalize outside of its training sample. For this task, our training dataset is not representative of the state of Alaska as a whole. Figure 5 compares the distribution of seven features between the training dataset and the in-AoA grid. These features were selected by hand to illustrate several known biases in the Alaska Permafrost Thaw Database, in addition to the geographic clustering shown in Figure 2 (Webb et al., 2026). Training points are flatter (median slope of 0.74 versus 3.4 degrees), closer to the nearest drainage (median height of 1 versus 15 m), and overall lower elevation (median height of 61 versus 260 m) than the rest of the state. They are much more likely to be open water (43% versus 3.8%) or emergent herbaceous wetlands (7.6% versus 3.9%), and much less likely to be deciduous forest (1% versus 3.7%), among other land cover indicators. This divergence motivates our decision to elevate area-of-applicability as a first-class model output alongside our log-evidence index, as well as a need for careful spatial cross-validation.

To evaluate how effectively the model can generalize outside of this geographically biased sample, we use spatial cross-validation, which holds out entire blocks of terrain so that nearby points cannot straddle the test-train splits. Figure 4a shows the full precision-recall curve for the operative 10 km block size, pooled across all splits, testing out-of-sample model skill. The model reaches an area under the precision-recall curve (AUC-

PR) of 0.852±0.011 across 20 reshuffles of the block assignments, which is approximately fifteen times greater than the floor associated with random chance. The area under the receiver operating characteristic (ROC) curve, which measures false positives against true positives, is 0.98 over the same folds. However, because ROC is insensitive the rare class, we do not use it as our primary metric.

Our baseline regression model recovers most of this performance. Logistic regression on the same folds achieves an AUC-PR of 0.776±0.017 (Figure 4a). We view this as the metric floor that a model must clear in order to provide useful predictive capability. The gradient-boosted model outperforms logistic regression across the full recall range, by a margin of 0.076 ± 0.019, stable across reshuffles.

Because the training points are concentrated regionally, we further investigate the model's robustness across space. Figure 4b tracks the AUC-PR against how far away the nearest training point typically lies from a given prediction. First, we construct block sizes of 5 to 200 km, corresponding to a median distance to the nearest training point of 1.8 to 30.8 km. To test larger distances, we cluster points using a k-means clustering algorithm on location only, producing between 3 and 50 larger approximate geographic regions, and perform the same hold-out analysis. The two methods overlap in the 20 to 30 km operational range, indicating an underlying pattern in model skill that is relatively insensitive to our choice in hold-out method. Skill decays gently from 0.88 when generalizing at a few kilometers to 0.54 at a median distance of 251 km. This AUC-PR would indicate a poor classifier on a balanced problem, but for our dataset is still roughly nine times greater than the chance floor. The predictions therefore survive extrapolation across hundreds of kilometers, rather than relying on a model that memorizes the feature space of specific neighborhoods.

Additional controls confirm the signal is robust rather than an artifact of our software pipeline. Randomly permuting the training labels collapses skill to chance, giving an AUC-PR of 0.062 against the 0.057 floor. Trivial baseline models score at the floor. Only four pairs of training points share identical features while disagreeing on their label, so label noise imparts no meaningful ceiling on separation. Longitude and latitude are excluded from the feature table, so the model does not ever use location directly as a predictor.

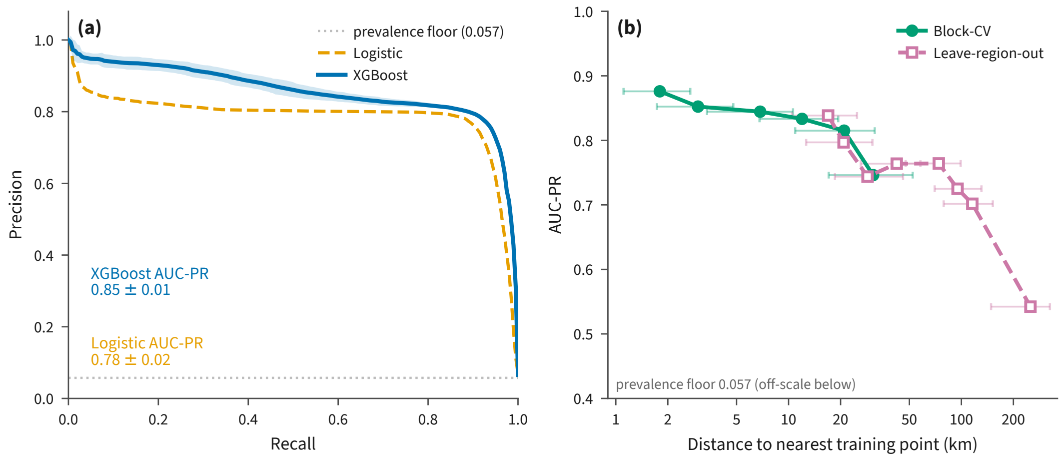


Figure 4.

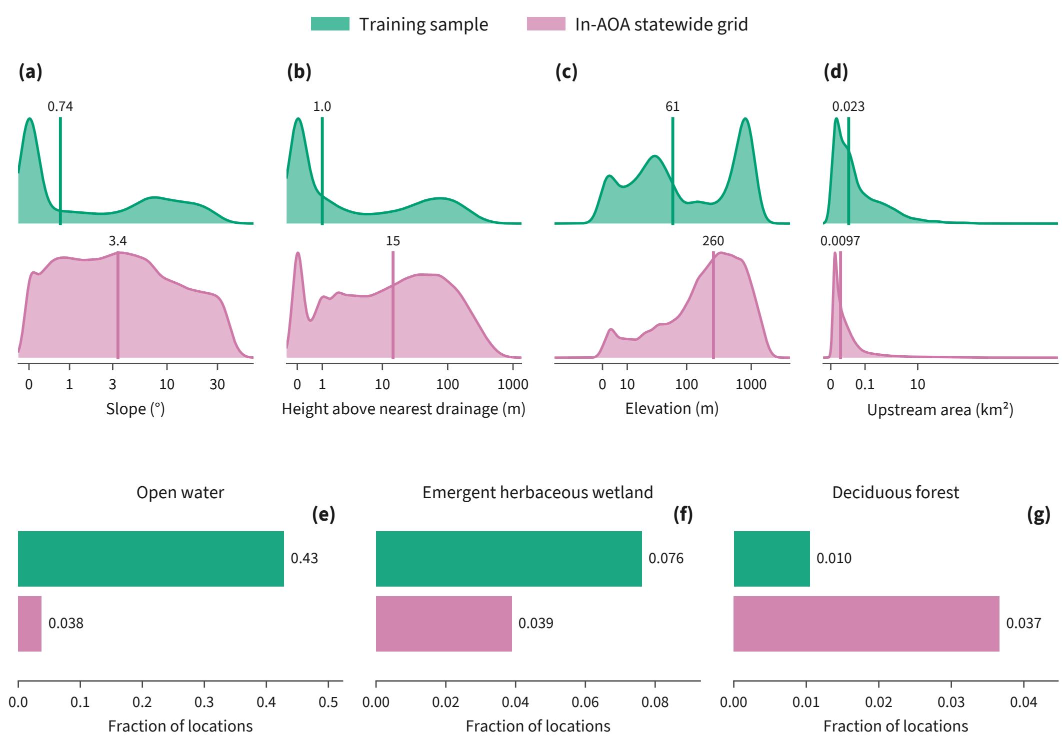


Figure 5.

4.3 Key Indicators of Abrupt Thaw

5 Discussion

6 Conclusion

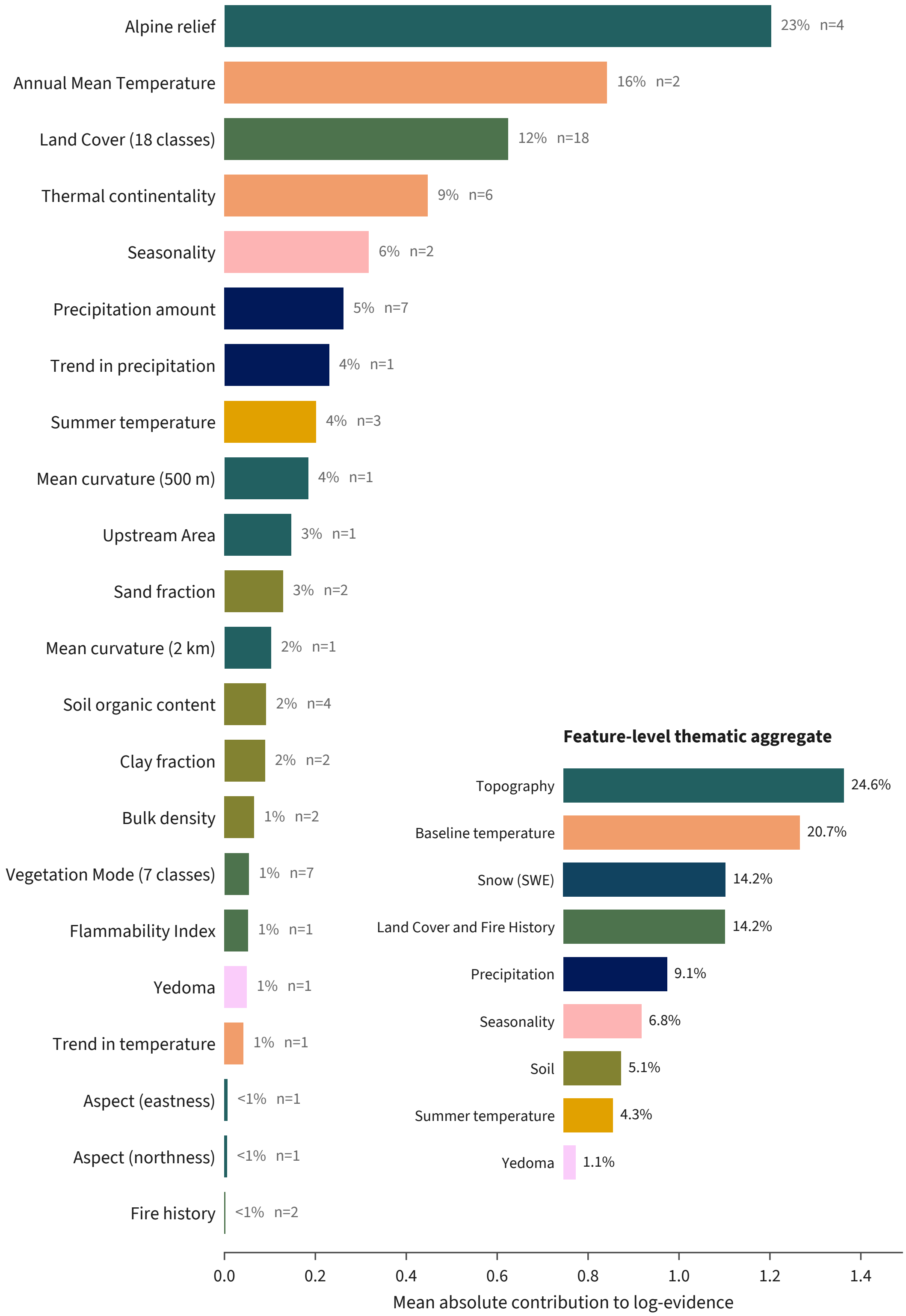
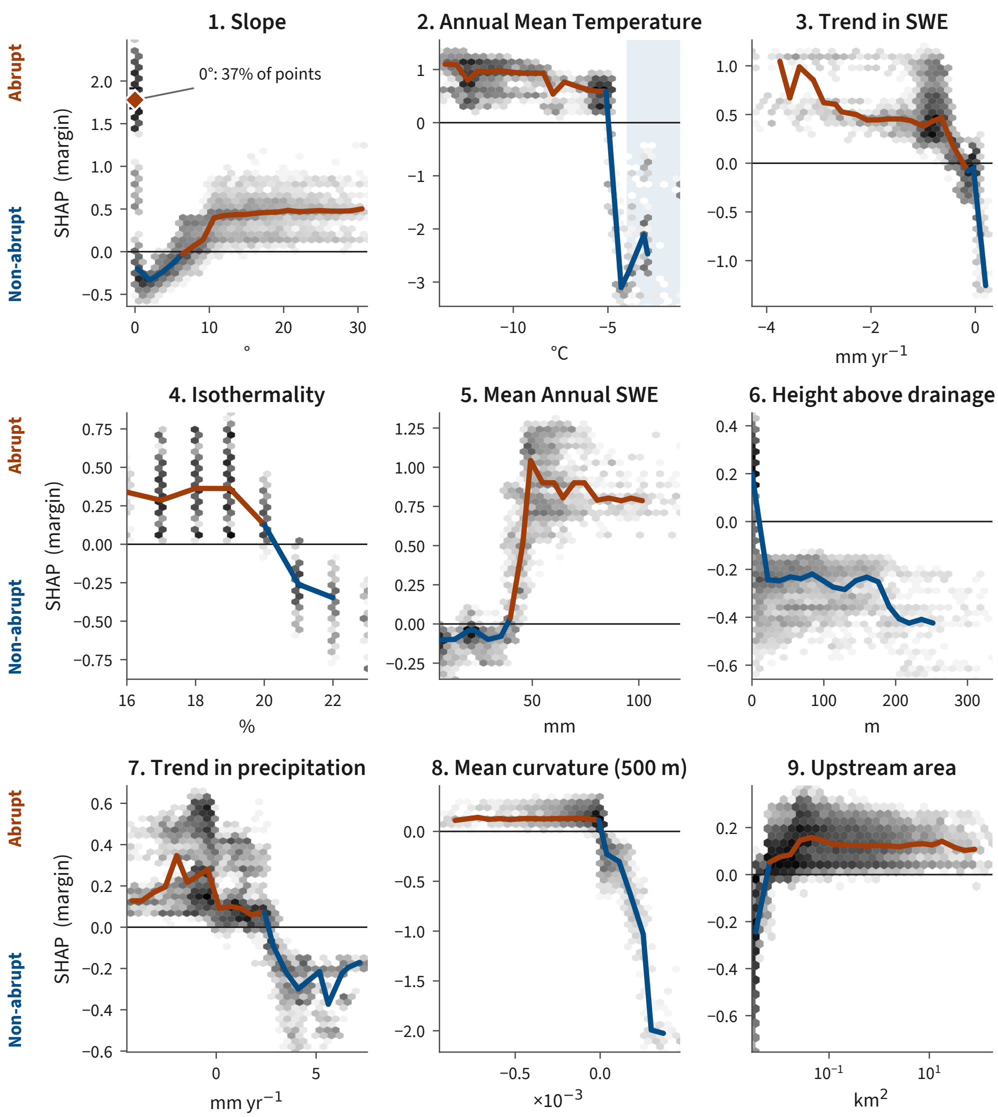

Alpine relief
23% n=4
Annual Mean Temperature
16% n=2
Land Cover (18 classes)
12% n=18
Thermal continentality
9% n=6
Seasonality
6% n=2
Precipitation amount
5% n=7
Trend in precipitation
4% n=1
Summer temperature
4% n=3
Mean curvature (500 m)
4% n=1
Upstream Area
3% n=1
Sand fraction
3% n=2
Mean curvature (2 km)
2% n=1
Soil organic content
2% n=4
Clay fraction
2% n=2
Bulk density
1% n=2
Vegetation Mode (7 classes)
1% n=7
Flammability Index
1% n=1
Yedoma
1% n=1
Trend in temperature
1% n=1
Aspect (eastness)
<1% n=1
Aspect (northness)
<1% n=1
Fire history
<1% n=2
Feature-level thematic aggregate
Topography
24.6%
Baseline temperature
20.7%
Snow (SWE)
14.2%
Land Cover and Fire History
14.2%
Precipitation
9.1%
Seasonality
6.8%
Soil
5.1%
Summer temperature
4.3%
Yedoma
1.1%
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Mean absolute contribution to log-evidence

Figure 6.



1. Slope
2. Annual Mean Temperature
3. Trend in SWE
4. Isothermality
5. Mean Annual SWE
6. Height above drainage
7. Trend in precipitation
8. Mean curvature (500 m)
9. Upstream area
Abrupt
Non-abrupt
SHAP (margin)
0°: 37% of points
°
°C
mm yr^{-1}
%
mm
m
×10^{-3}
km^2

Figure 7.

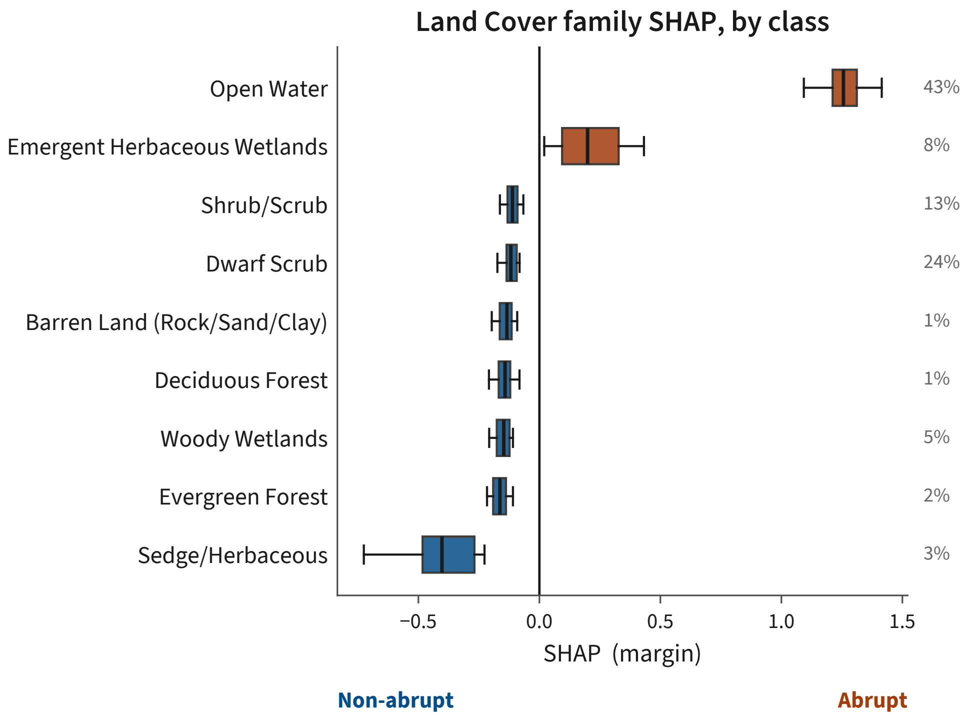


Figure 8.

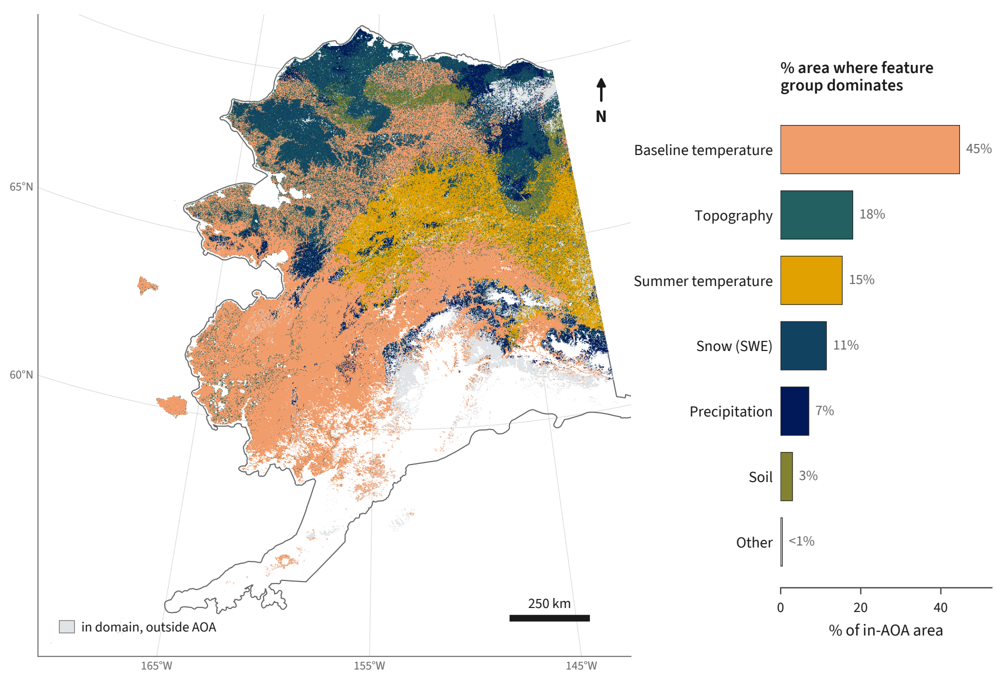


Figure 9.

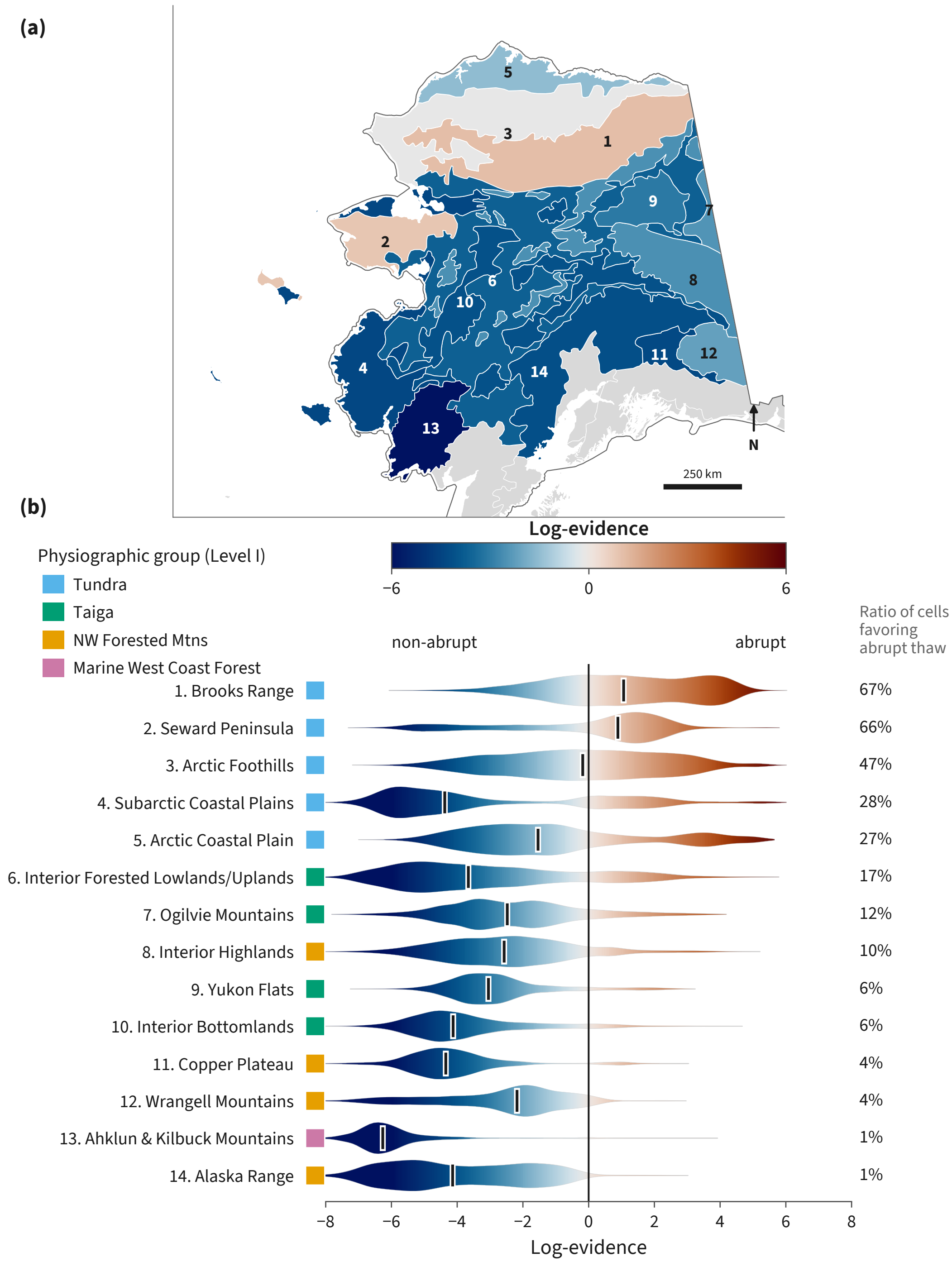
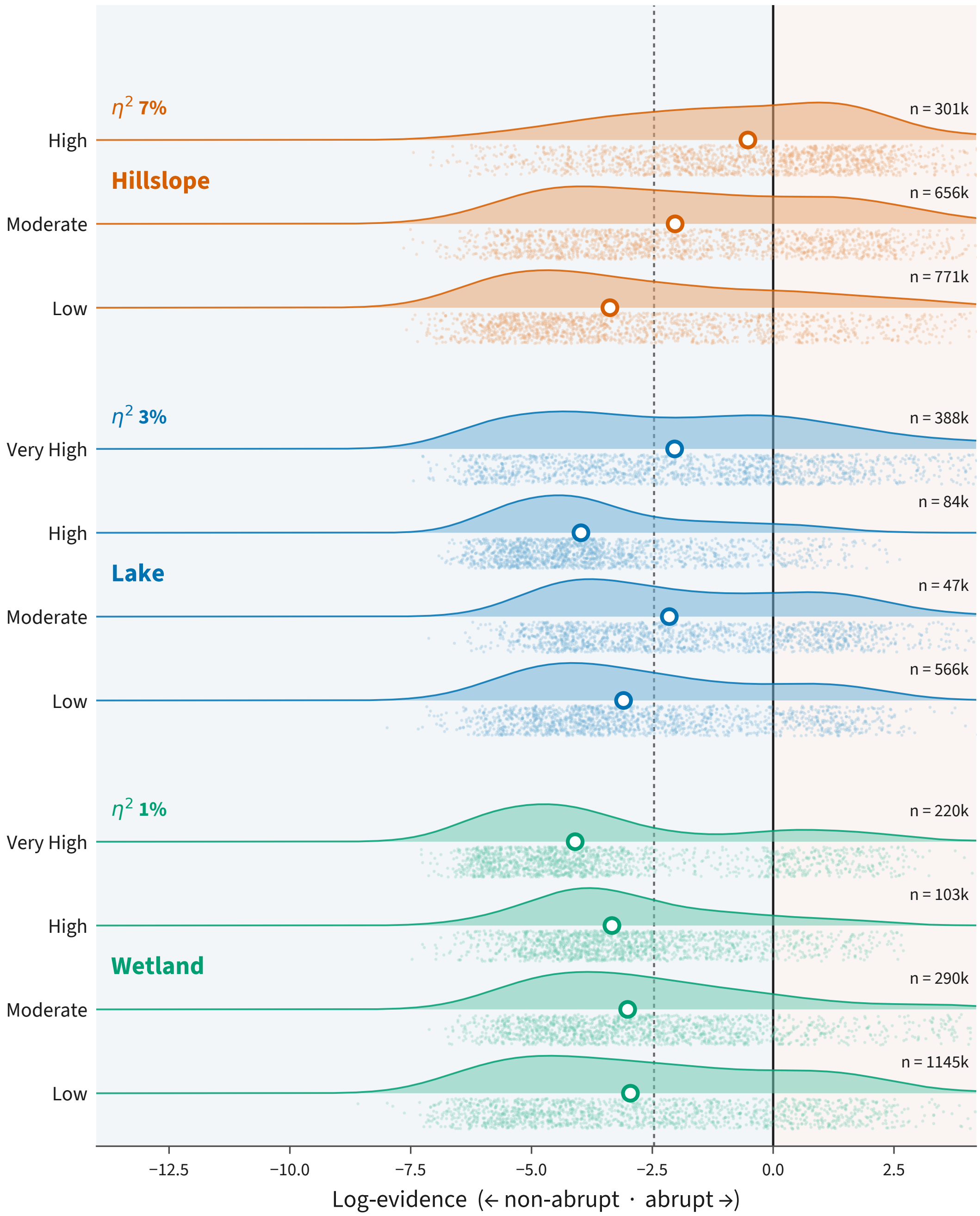

(a)
5
3
1
9
7
2
8
6
10
11
12
4
14
13
N
250 km
(b)
Log-evidence
−6
0
6
Physiographic group (Level I)
Tundra
Taiga
NW Forested Mtns
Marine West Coast Forest
non-abrupt
abrupt
Ratio of cells favoring abrupt thaw
1. Brooks Range 67%
2. Seward Peninsula 66%
3. Arctic Foothills 47%
4. Subarctic Coastal Plains 28%
5. Arctic Coastal Plain 27%
6. Interior Forested Lowlands/Uplands 17%
7. Ogilvie Mountains 12%
8. Interior Highlands 10%
9. Yukon Flats 6%
10. Interior Bottomlands 6%
11. Copper Plateau 4%
12. Wrangell Mountains 4%
13. Ahklun & Kilbuck Mountains 1%
14. Alaska Range 1%
−8
−6
−4
−2
0
2
4
6
8
Log-evidence

Figure 10.



η² 7%
High
Hillslope
Moderate
Low
n = 301k
n = 656k
n = 771k
η² 3%
Very High
High
Lake
Moderate
Low
n = 388k
n = 84k
n = 47k
n = 566k
η² 1%
Very High
High
Wetland
Moderate
Low
n = 220k
n = 103k
n = 290k
n = 1145k
−12.5
−10.0
−7.5
−5.0
−2.5
0.0
2.5
Log-evidence (← non-abrupt · abrupt →)

Figure 11.

Open Research Statement

Statement text.

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Appendix A Data sources

Appendix B Hyperparameters

Appendix C Spatial buffer

Appendix D Area-of-applicability

Appendix E SHAP feature families

Table A1. Geospatial data products used to build the feature set. "Cols" gives the number of columns each product contributes to the 70-column model matrix, after one-hot encoding of the categorical land-cover and vegetation-mode sources and depth aggregation of the soil variables.

Data product	Native res.	Variables derived	Cols	Reference
Remote sensing				
USGS 3DEP DEM	10 m	Elevation, slope, aspect (northness, eastness), mean curvature (500 m, 2 km)	6	(U.S. Geological Survey, 2019)
MERIT Hydro v1.0.1	~90 m	Height above nearest drainage, upstream area	2	(Yamazaki et al., 2019)
NLCD 2016 (Alaska)	30 m	Land cover (one-hot)	18	(Dewitz, 2019)
MODIS MCD64A1	~500 m	Time since last fire, burn count	2	(Giglio et al., 2018)
Community and modeled data products				
SoilGrids (ISRIC)	250 m	Organic carbon, nitrogen, bulk density, sand, clay (two depths)	10	(Poggio et al., 2021)
ALFRESCO (SNAP)	~1 km	Flammability, vegetation mode (one-hot)	8	(Bennett et al., 2017)
IRYP v2 (yedoma)	Vector	Confirmed yedoma presence	1	(Strauss et al., 2022)
Gridded climate				
WorldClim v1	~1 km	19 bioclimatic variables	19	(Hijmans et al., 2005)
Daymet V4	1 km	Mean annual SWE, and SWE, precipitation, and temperature trends (1991–2020)	4	(Thornton et al., 2022)
Total			70	